Single-Snapshot Localization Using Sparse Extremely Large Aperture Arrays

Yunqiao Hu†, Xuesu Xiao§, Steven Jones†, and Shunqiao Sun†
†University of Alabama, Tuscaloosa, AL 35487 USA
§George Mason University, Fairfax, VA 22030 USA

***Abstract*—This paper investigates single-snapshot direction-of-arrival (DOA) estimation and target localization with coherent sparse extremely large aperture arrays (ELAAs) in automotive radar applications. Far-field and near-field signal models are formulated for distributed bistatic configurations. To enable noncoherent processing, a single-snapshot MUSIC (SS-MUSIC) algorithm is proposed to fuse local spectra from individual subarrays and extended to near-field localization via geometric intersection. For coherent processing, a single-snapshot ESPRIT (SS-ESPRIT) method with ambiguity dealiasing is developed to fully exploit the aperture of sparse ELAAs for high-resolution angle estimation. Simulation results demonstrate that SS-ESPRIT provides superior angular resolution for closely spaced far-field targets, while SS-MUSIC offers robustness in near-field localization and flexibility in hybrid scenarios.**

***Index Terms*—Extremely large aperture arrays, Direction-of-arrival estimation, Single-snapshot, Far-field, Near-field**

I. Introduction

Sparse extremely large aperture arrays (ELAAs) have attracted increasing interest in communications and radar sensing because they synthesize very large effective apertures using spatially distributed antennas [1], [2]. Compared with conventional uniform linear arrays (ULAs), sparse ELAAs provide higher angular resolution, greater spatial diversity, and wider field of view (FoV) with reduced hardware cost. Their bi-/multi-static configurations also enable coherent processing across widely separated nodes, improving detection and localization performance.

In wireless communications, sparse ELAAs have been studied under the extremely large-scale MIMO (XL-MIMO) paradigm for beyond-5G/6G systems [2]–[4]. In radar sensing, related distributed aperture radar architectures have been investigated for early warning, surveillance, and networked sensing [5]–[8], typically using centralized or coherent processing to improve signal-to-noise ratio (SNR) and localization accuracy. More recently, distributed radar has been extended to automotive applications [9], [10], where multiple radar modules mounted on a vehicle form a sparse ELAA capable of both far-field and near-field direction-of-arrival (DOA) estimation. This spatial diversity supports multi-view sensing and improves the detection of targets with angle-dependent radar cross sections (RCS) [11]–[13]. Industrial systems such as Zendar's distributed aperture radar [14] further highlight the practical potential of this architecture.

Despite these advantages, single-snapshot DOA estimation and target localization with sparse ELAAs remain challenging in automotive scenarios. Most existing methods rely on multiple snapshots to estimate the signal subspace [15], [16], which limits their applicability in highly dynamic environments where only one snapshot may be available [17]–[19]. In addition, large inter-module spacing introduces strong sidelobes and spatial ambiguities, while the large synthesized aperture makes near-field effects non-negligible. Although near-field sensing for large-aperture arrays has been studied in [20], [21], sparse-ELAA-specific single-snapshot localization remains insufficiently explored.

In this paper, we address this problem by developing single-snapshot far-field and near-field signal models for sparse bistatic ELAAs, together with two complementary estimation frameworks. We propose a noncoherent single-snapshot MUSIC (SS-MUSIC) method that fuses local subarray spectra for robust estimation and extends naturally to near-field localization through association and geometric intersection. We also propose a coherent single-snapshot ESPRIT (SS-ESPRIT) method with ambiguity dealiasing for high-resolution far-field DOA estimation. Simulation results show that SS-ESPRIT achieves superior far-field resolution, especially for closely spaced targets, whereas SS-MUSIC provides robust near-field localization performance.

II. System Model

A. Modeling Assumptions

We consider a synchronized sparse ELAA with orthogonal TDM-MIMO transmission, where each radar transmits sequentially while all nodes receive simultaneously [9], [17]. As shown in Fig. 1, the ELAA consists of two M-element ULAs with inter-element spacing $d = \lambda/2$ and inter-module spacing $D_s \gg (M-1)d$, yielding a total aperture $D_a = D_s + 2(M-1)d$. Let $m \in \{0, \ldots, M-1\}$ and $n \in \{1, 2\}$ index the element and ULA, respectively. The position of the (n, m)-th element is $\mathbf{u}_{n,m} = [x_{n,m}, 0]^T$, with

$$x_{n,m} = \left(m - \tfrac{M-1}{2}\right)d + \left(n - \tfrac{3}{2}\right)\left(D_s + (M-1)d\right). \quad (1)$$

Assume K targets located at $\mathbf{p}_k = r_k[\sin\theta_k, \cos\theta_k]^T$ ($k = 1, \ldots, K$), where r_k and θ_k are the range and DOA relative to

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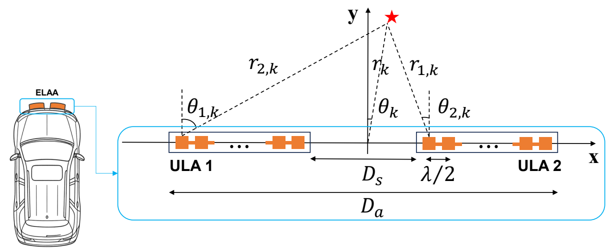


Fig. 1: Illustration of a sparse ELAA composed of two uniform linear arrays (ULAs) symmetrically placed along the x-axis.

the ELAA center. The (n,m)-th element of the steering vector is

$$\begin{aligned} a_{n,m}(r_k,\theta_k) &= \exp\left(-j\tfrac{2\pi}{\lambda}\left\|\mathbf{p}_k-\mathbf{u}_{n,m}\right\|\right) \\ &= \exp\left(-j\tfrac{2\pi}{\lambda}\sqrt{r_k^2-2r_kx_{n,m}\sin\theta_k+x_{n,m}^2}\right). \end{aligned} \tag{2}$$

To perform accurate DOA estimation, far-field and near-field regimes must be distinguished. When $r_k \geq 2D_a^2/\lambda$, the planar wavefront assumption allows the propagation distance to the (n,m)-th sensor to be approximated as $r_{n,m}^k \approx r_k - x_{n,m}\sin\theta_k$ simplifying (2) to:

$$a_{n,m}(r_k,\theta_k) \approx e^{j\frac{2\pi}{\lambda}x_{n,m}\sin\theta_k}, \quad n\in\mathcal{N},\ m\in\mathcal{M}, \tag{3}$$

where the common range-dependent phase term is omitted. For $\frac{2S_a^2}{\lambda} \leq r_k < \frac{2D_a^2}{\lambda}$ with $S_a = (M-1)d$ denoting the aperture of each local ULA, the target lies in the near-field of the global ELAA but remains in the far-field of each local ULA. Let $r_{n,0}^k$ denote the distance from the k-th target to the reference element ($m=0$) of the n-th ULA:

$$r_{n,0}^k = \sqrt{r_k^2-2r_kx_{n,0}\sin\theta_k+x_{n,0}^2}, \forall n\in\mathcal{N}, \tag{4}$$

where $x_{n,0} = (n-2)(M-1)d + (n-\frac{3}{2})D_s$. The steering element is approximated by

$$a_{n,m}(r_k,\theta_k) \approx e^{-j\frac{2\pi}{\lambda}r_{n,0}^k}\, e^{j\frac{2\pi}{\lambda}md\sin\theta_{n,k}}, \quad n\in\mathcal{N},\ m\in\mathcal{M}, \tag{5}$$

where $\theta_{n,k}$ represents the DOA with respect to the n-th ULA. When

$$\max\left(5D_a, \tfrac{4D_aD_s}{\lambda}\right) \leq r_k < \tfrac{2D_a^2}{\lambda}, \tag{6}$$

the DOA variation across ULAs becomes negligible [22], [23], so $\theta_{n,k} \approx \theta_k$ and (5) simplifies to

$$a_{n,m}(r_k,\theta_k) = e^{-j\frac{2\pi}{\lambda}r_{n,0}^k} \cdot e^{j\frac{2\pi}{\lambda}md\sin\theta_k}, \quad n\in\mathcal{N},\ m\in\mathcal{M}. \tag{7}$$

The single-snapshot received signal is modeled as

$$\mathbf{x} = \sum_{k=1}^{K} s_k\mathbf{a}(r_k,\theta_k) + \mathbf{n}, \tag{8}$$

where s_k denotes the RCS coefficient and $\mathbf{n}$ is additive white Gaussian noise.

III. Localization with Sparse ELAAs

This section investigates single-snapshot localization with sparse ELAAs under both far-field and near-field conditions.

A. Far-Field DOA Estimation with Sparse ELAAs

Large inter-module spacing improves angular resolution but also produces strong sidelobes, as shown in Fig. 2. We

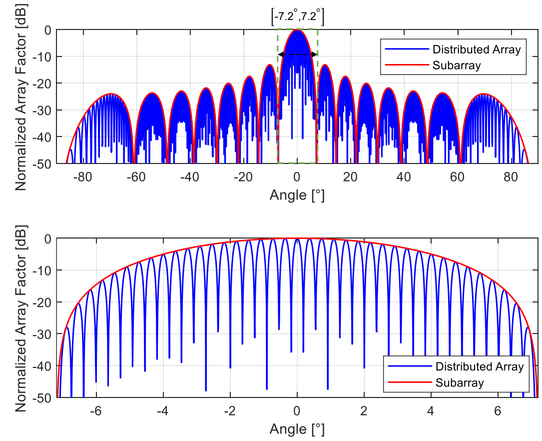


Fig. 2: Array factor of sparse ELAA vs. sub-ULA ($D_s = 0.59$ m, $M=16$). Bottom: Zoomed-in view of $[-7.2°, 7.2°]$.

propose two single-snapshot far-field DOA estimation methods based on SS-MUSIC [24] and SS-ESPRIT [25]. Let $\mathbf{y}_n = [\mathbf{y}_{n,1}; \mathbf{y}_{n,2}]$ denote the measurements from the two ULAs. For each ULA, a Hankel matrix $\mathcal{H}(\mathbf{y}_{n,l}) \in \mathbb{C}^{(L+1)\times(M-L+1)}$ is constructed according to (9), where l denotes the radar node. When $K<M$, $\mathcal{H}(\mathbf{y}_{n,l})$ is low rank. $\mathcal{H}(\mathbf{y}_{n,l})$ is defined as:

$$\mathcal{H}(\mathbf{y}_{n,l}) = \begin{bmatrix} \mathbf{y}_{n,l}(0) & \mathbf{y}_{n,l}(1) & \cdots & \mathbf{y}_{n,l}(M-L) \\ \mathbf{y}_{n,l}(1) & \mathbf{y}_{n,l}(2) & \cdots & \mathbf{y}_{n,l}(M-L+1) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{y}_{n,l}(L) & \mathbf{y}_{n,l}(L+1) & \cdots & \mathbf{y}_{n,l}(M) \end{bmatrix},$$
$$l\in\{1,2\}, 1\leq L<M \tag{9}$$

Applying Singular Value Decomposition (SVD) yields:

$$\mathcal{H}(\mathbf{y}_{n,l}) = \widetilde{\mathbf{U}}_l\widetilde{\boldsymbol{\Sigma}}_l\widetilde{\mathbf{V}}_l^H, \tag{10}$$

where $\widetilde{\mathbf{U}}_l \in \mathbb{C}^{(L+1)\times q}$, $\widetilde{\boldsymbol{\Sigma}}_l \in \mathbb{C}^{q\times q}$, and $\widetilde{\mathbf{V}}_l \in \mathbb{C}^{(M-L+1)\times q}$. Partitioning the singular vectors as

$$\widetilde{\mathbf{U}}_l = \begin{bmatrix}\mathbf{U}_{s,l} & \mathbf{U}_{n,l}\end{bmatrix}, \qquad \widetilde{\mathbf{V}}_l = \begin{bmatrix}\mathbf{V}_{s,l} & \mathbf{V}_{n,l}\end{bmatrix}, \tag{11}$$

with $\mathbf{U}_{s,l} \in \mathbb{C}^{(L+1)\times K}$, $\mathbf{U}_{n,l} \in \mathbb{C}^{(L+1)\times(q-K)}$, $\mathbf{V}_{s,l} \in \mathbb{C}^{(M-L+1)\times K}$, and $\mathbf{V}_{n,l} \in \mathbb{C}^{(M-L+1)\times(q-K)}$, the columns of $\mathbf{U}_{s,l}$ and $\mathbf{U}_{n,l}$ span the signal and noise subspaces, respectively.

$$S_l(\theta) = \frac{\|\mathbf{A}_l(\theta)\|_2}{\|\mathbf{U}_{n,l}^H\mathbf{A}_l(\theta)\|_2}, \quad l=1,2 \tag{12}$$

and outputs $\hat{S}(\theta) = \max\{S_l(\theta)\}$.

SS-MUSIC processes each ULA separately and combines estimates noncoherently, failing to fully utilize the ELAA's large aperture. To improve resolution, we develop SS-ESPRIT by exploiting the shift-invariant structure of the ELAA. Two identical subarrays are selected by matrices $\mathbf{J}_1$ and $\mathbf{J}_2$ such that:

$$\mathbf{A}_{s1}(\boldsymbol{\theta}) = \mathbf{J}_1\mathbf{A}_s(\boldsymbol{\theta}), \quad \mathbf{A}_{s2}(\boldsymbol{\theta}) = \mathbf{J}_2\mathbf{A}_s(\boldsymbol{\theta}), \tag{13}$$

with shift-invariance:

$$\mathbf{J}_1\mathbf{A}_s(\boldsymbol{\theta})\boldsymbol{\Phi} = \mathbf{J}_2\mathbf{A}_s(\boldsymbol{\theta}), \tag{14}$$

where $\boldsymbol{\Phi} = \mathrm{diag}(e^{j\phi_1}, e^{j\phi_2}, ..., e^{j\phi_K})$ with $\phi_i = 2\pi\frac{\Delta_s}{\lambda}\sin(\theta_i)$, and Δ_s is the shift length between two subarrays. The signal subspaces $\mathbf{U}_{s,1}$ and $\mathbf{U}_{s,2}$ are obtained via (9)–(12) and concatenated to subspace $\mathbf{U}_s = [\mathbf{U}_{s,1}; \mathbf{U}_{s,2}] \in \mathbb{C}^{2(L+1)\times K}$. Assuming a nonsingular transformation $\mathbf{T}_a \in \mathbb{C}^{K\times K}$, the signal subspace $\mathbf{U}_s$ and the array steering matrix $\mathbf{A}_s(\boldsymbol{\theta})$ satisfy:

$$\mathbf{U}_s = \mathbf{A}_s(\boldsymbol{\theta})\mathbf{T}_a. \tag{15}$$

Substituting (15) into the shift-invariance relation in (14) yields:

$$\mathbf{J}_1\mathbf{U}_s\boldsymbol{\Psi} = \mathbf{J}_2\mathbf{U}_s, \tag{16}$$

where $\boldsymbol{\Psi} = \mathbf{T}_a^{-1}\boldsymbol{\Phi}\mathbf{T}_a \in \mathbb{C}^{K\times K}$. $\boldsymbol{\Psi}$ and $\boldsymbol{\Phi}$ are similar matrices and share the same eigenvalues. With the definitions $\mathbf{U}_{1s} = \mathbf{J}_1\mathbf{U}_s, \mathbf{U}_{2s} = \mathbf{J}_2\mathbf{U}_s$, (16) is solved via the least squares (LS) method:

$$\hat{\boldsymbol{\Psi}} = (\mathbf{U}_{1s}^H\mathbf{U}_{1s})^{-1}\mathbf{U}_{1s}^H\mathbf{U}_{2s}. \tag{17}$$

The eigenvalues of $\hat{\boldsymbol{\Psi}}$ are denoted by $\{\hat{\xi}_i\}$. Let $\hat{\phi}_i = \mathrm{Arg}(\hat{\xi}_i) \in (-\pi, \pi]$. The unaliased DOA estimate is

$$\hat{\theta}_i = \arcsin\left(\frac{\lambda\hat{\phi}_i}{2\pi\Delta_s}\right). \tag{18}$$

When $\Delta_s > \lambda/2$, aliasing occurs and the DOA candidates are

$$\hat{\theta}_i^{(q)} = \arcsin\left(\frac{\lambda}{2\pi\Delta_s}(\hat{\phi}_i + 2\pi q)\right), \tag{19}$$

where q satisfies

$$\left\lceil -\frac{\Delta_s}{\lambda} - \frac{\hat{\phi}_i}{2\pi} \right\rceil \le q \le \left\lfloor \frac{\Delta_s}{\lambda} - \frac{\hat{\phi}_i}{2\pi} \right\rfloor, \quad q \in \mathbb{Z}. \tag{20}$$

Here $\lceil\cdot\rceil$ and $\lfloor\cdot\rfloor$ denote the ceiling and floor functions, respectively. Unambiguous DOAs are then resolved by checking consistency across varying Δ_s, as outlined in Algorithm 1. We assume the number of targets K is known; in practice, it can be estimated from the Hankel singular values using standard criteria such as AIC, MDL [26], or singular-value-gap thresholding.

B. Near-Field Localization with Sparse ELAAs

For targets located in the near-field of the global ELAA ($\frac{2S_a^2}{\lambda} \le r_k < \frac{2D_a^2}{\lambda}$), the apparent DOAs observed by different ULAs are no longer identical and follow the near-field steering model in (5). However, each local ULA still satisfies the far-field condition, allowing SS-MUSIC to be applied independently to estimate DOAs from each subarray. Thus, the DOA estimates must be associated across ULAs. We employ the Matching Pursuit algorithm [27] to pair corresponding estimates. Given an associated pair $(\hat{\theta}_{1,k}, \hat{\theta}_{2,k})$, and known subarray reference positions $\mathbf{c}_1$ and $\mathbf{c}_2$, the bearing vectors from each subarray are:

$$\mathbf{d}_i = [\sin(\hat{\theta}_{i,k}), \cos(\hat{\theta}_{i,k})]^T, \quad i = 1, 2.$$

The target locations are obtained by the least-squares intersection of the two bearing lines $\ell_1(r_{1,k}) = \mathbf{c}_1 + r_{1,k}\mathbf{d}_1$ and $\ell_2(r_{2,k}) = \mathbf{c}_2 + r_{2,k}\mathbf{d}_2$. The near-field localization is summarized in Algorithm 2.

Algorithm 1 SS-ESPRIT for Far-Field DOA Estimation with ELAA

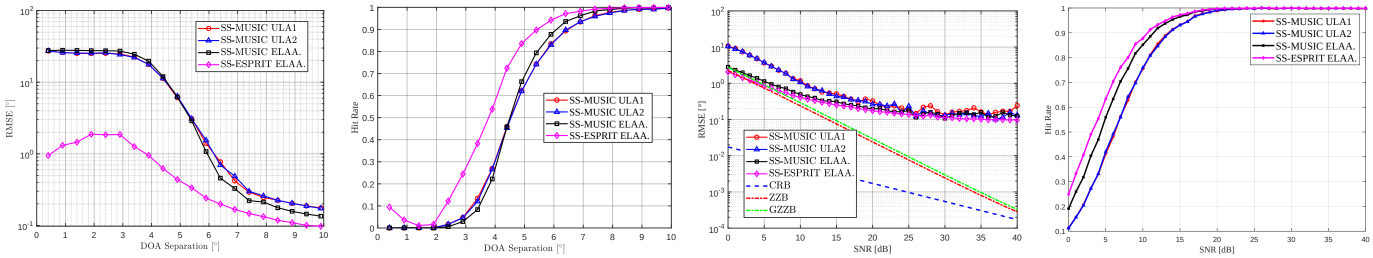
Input: Measurement vector $\mathbf{y}_n = [\mathbf{y}_{n,1}; \mathbf{y}_{n,2}] \in \mathbb{C}^{2M\times 1}$
Construct Hankel matrices $\mathcal{H}(\mathbf{y}_{n,1})$ and $\mathcal{H}(\mathbf{y}_{n,2}) \in \mathbb{C}^{(L+1)\times(M-L+1)}$
Perform SVD on each Hankel matrix to extract signal subspaces $\mathbf{U}_{s,1}$ and $\mathbf{U}_{s,2}$
Form the concatenated signal subspace: $\mathbf{U}_s = [\mathbf{U}_{s,1}; \mathbf{U}_{s,2}] \in \mathbb{C}^{2(L+1)\times K}$
for each subarray shift u **do**
    Select subspaces $[\mathbf{U}_{1s}^{(u)}, \mathbf{U}_{2s}^{(u)}]$ using selection matrices $\mathbf{J}_1^{(u)}$, $\mathbf{J}_2^{(u)}$
    Estimate $\hat{\boldsymbol{\Psi}}^{(u)}$ via (17) and compute DOA candidates using (18) and (19)
end for
Apply consistency checks to candidate sets to resolve final DOAs.

Algorithm 2 SS-MUSIC for Near-Field DOA and Target Localization

Input: Measurement vector $\mathbf{y}_n = [\mathbf{y}_{n,1}; \mathbf{y}_{n,2}] \in \mathbb{C}^{2M\times 1}$
Construct Hankel matrices $\mathcal{H}(\mathbf{y}_{n,1})$ and $\mathcal{H}(\mathbf{y}_{n,2})$
Apply SVD to extract noise subspaces $\mathbf{U}_{n,1}$ and $\mathbf{U}_{n,2}$
Compute SS-MUSIC spectra and estimate local DOAs $\hat{\boldsymbol{\theta}}_1$, $\hat{\boldsymbol{\theta}}_2$
Associate $\hat{\boldsymbol{\theta}}_1$, $\hat{\boldsymbol{\theta}}_2$ using Matching Pursuit
Estimate target positions by intersecting bearing lines from each ULA
Output: Estimated target positions $[\hat{\mathbf{x}}_{\text{target}}, \hat{\mathbf{y}}_{\text{target}}]$

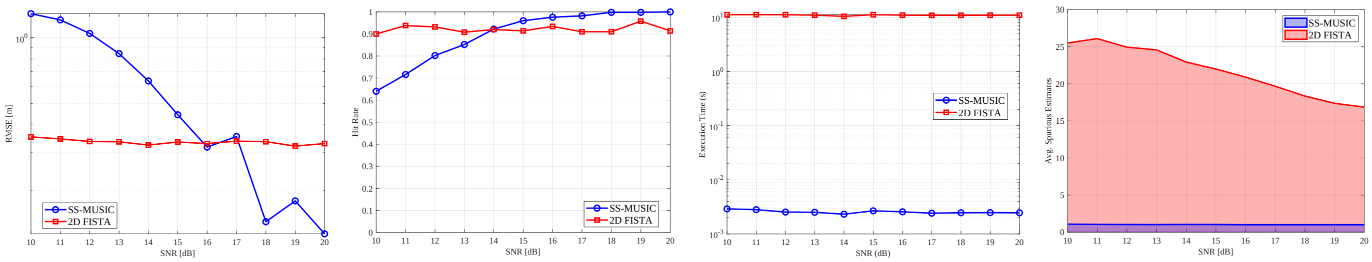
IV. Numerical Results

Far-field DOA performance is evaluated using a $76\,\mathrm{GHz}$ sparse ELAA consisting of two 16-element ULAs ($d = \lambda/2$) separated by 150λ ($D_s \approx 0.59\,\mathrm{m}$). With a Fraunhofer distance of approximately $215\,\mathrm{m}$, targets are placed at $r = 250\,\mathrm{m}$. We perform $5{,}000$ Monte Carlo trials with DOAs uniformly drawn from $[-60°, 60°]$ under two scenarios: (i) varying angular separation $[0.4°, 10°]$ at $15\,\mathrm{dB}$ SNR, and (ii) varying SNR $[0, 40]\,\mathrm{dB}$ at a fixed $10°$ separation. The proposed SS-ESPRIT is compared with SS-MUSIC (applied to both the sparse array and individual ULAs) and theoretical bounds (CRB [28], ZZB [29], GZZB [30]). Performance is assessed via the Hit



(a) (b) (c) (d)

Fig. 3: Performance of SS-MUSIC and SS-ESPRIT with ELAA: (a) RMSE, SNR=15dB; (b) Hit rate, SNR=15dB; (c) RMSE, 10° separation; (d) Hit rate, 10° separation.



(a) (b) (c) (d)

Fig. 4: Performance comparison of the proposed SS-MUSIC and 2D FISTA in near-field scenarios vs. SNR. The metrics evaluated are (a) RMSE of localization, (b) Hit rate, (c) Average execution time, and (d) The average count of spurious targets.

Rate (percentage of estimates within $\pm 0.5°$) and Root Mean Squared Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{KN_{\mathrm{trial}}} \sum_{n=1}^{N_{\mathrm{trial}}} \sum_{k=1}^{K} \left(\hat{\theta}_k^{(n)} - \theta_k\right)^2}. \quad (21)$$

As shown in Fig. 3(a)–(b), SS-ESPRIT consistently achieves lower RMSE and higher hit rate than SS-MUSIC, with the advantage becoming more pronounced for closely spaced targets ($< 5°$). In the low-SNR regime (0–5 dB), Fig. 3(c)–(d) shows that SS-ESPRIT approaches the ZZB and GZZB more closely than the competing methods. The difference between SS-MUSIC and SS-ESPRIT is mainly due to how they use the estimated subspace: SS-MUSIC relies on a noise-subspace spectral search and is therefore more sensitive to sidelobe contamination and peak broadening, whereas SS-ESPRIT exploits rotational invariance and is more sensitive to subarray conditioning and invariance mismatch. Near-field localization is evaluated using two targets at $r = 5\,\mathrm{m}$, which lies within the near-field region $r < \max(5D_a, \frac{4D_aD_s}{\lambda}) \approx 19.5\,\mathrm{m}$. The DOAs are randomly drawn from $[-20°, 20°]$ with $10°$ separation. We perform 1,000 Monte Carlo trials over SNRs from 10 to 20 dB and compare SS-MUSIC with 2D FISTA [31]. Different SNR ranges are used across figures only to highlight the most informative transition regime in each experiment; all other simulation settings remain unchanged. We include 2D FISTA as a representative high-resolution optimization-based baseline for joint near-field localization on a discretized angle-range grid. In contrast, the proposed SS-MUSIC method performs per-subarray single-snapshot DOA estimation followed by cross-subarray association and geometric intersection. Localization accuracy is measured by

$$\mathrm{RMSE} = \sqrt{\frac{1}{KN_{\mathrm{trial}}} \sum_{n=1}^{N_{\mathrm{trial}}} \sum_{k=1}^{K} \left\| \hat{\mathbf{p}}_k^{(n)} - \mathbf{p}_k \right\|_2^2}. \quad (22)$$

As shown in Fig. 4(a), SS-MUSIC yields higher localization RMSE than 2D FISTA in the low-SNR regime (10–14 dB) because of its smaller effective aperture. However, at higher SNRs ($>$ 15 dB), SS-MUSIC achieves lower RMSE and substantially fewer spurious detections, while 2D FISTA maintains a high hit rate but is more prone to sidelobe-induced false alarms. As illustrated in Fig. 4(d), 2D FISTA retains high sidelobes that cause spurious detections even at high SNR, whereas SS-MUSIC produces cleaner spectra with fewer false alarms. In addition, SS-MUSIC is more than two orders of magnitude faster than 2D FISTA on an Intel Core i7-8700 CPU, making it attractive for real-time automotive radar.

V. Conclusions

This paper developed single-snapshot DOA estimation and localization methods for sparse distributed aperture radar in both far-field and near-field regimes. SS-ESPRIT exploits coherent processing over the synthesized aperture for high-resolution far-field DOA estimation, while SS-MUSIC operates on local subarrays and extends naturally to near-field localization through geometric intersection. Simulation results show that SS-ESPRIT provides better far-field resolution for closely spaced targets, whereas SS-MUSIC remains effective for near-field localization under high SNR. These results suggest that the proposed frameworks are promising for practical automotive radar applications.

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